

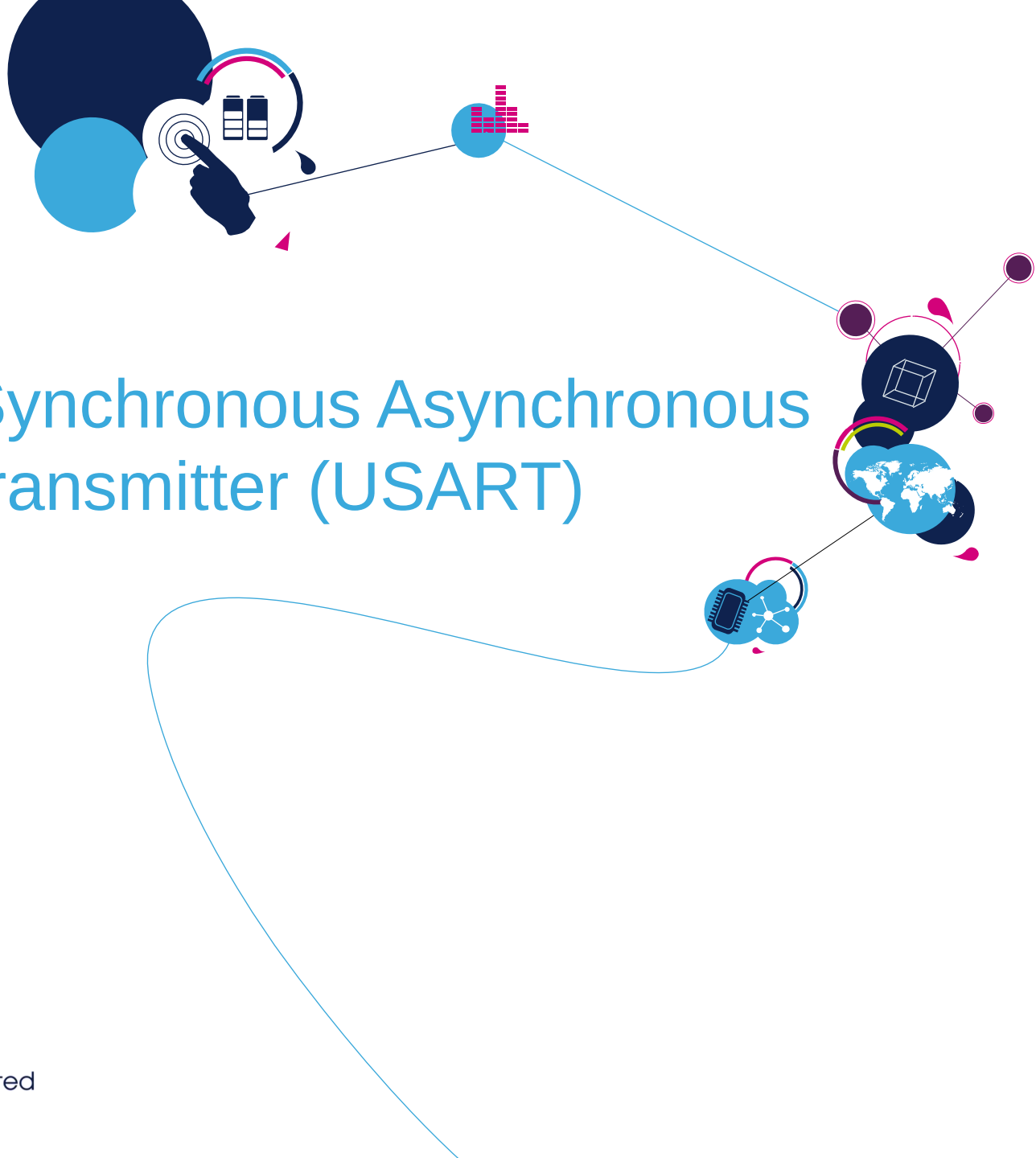


STM32F3 Technical Training

For reference only

Refer to the latest documents for details





Universal Synchronous Asynchronous Receiver Transmitter (USART)

USART Features (1/3)

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- Fully-programmable serial interface characteristics:
 - Data can be 8 or 9 bits
 - Even, odd or no-parity bit generation and detection
 - 1, 1.5 or 2 stop bit generation
 - Programmable baud rate generator
 - Configurable oversampling method by 16 or by 8
 - Up to 9Mbps when the clock frequency is 72 MHz and oversampling by 8 is selected.
- Programmable data order with MSB or LSB first.
- Swappable Tx/Rx pin configuration
- Tx/Rx pins active level inversion & Binary data inversion
- Support hardware flow control (CTS and RTS)
- Dual clock domain allowing
 - UART functionality and wakeup from Stop mode
 - Convenient baud rate programming independent from the PCLK reprogramming
- Dedicated transmission and reception flags (TxE and RxNE) with interrupt capability

USART Features (2/3)

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- Support for DMA
 - Receive DMA request
 - Transmit DMA request
- LIN Master compatible
- Synchronous Mode: Master mode only
- IrDA SIR Encoder Decoder
- Smartcard Capability T = 0, T = 1 (using the Address/character match, End of block, receiver timeout etc...)
- Basic support for Modbus communication (using Address/character match and Receiver timeout features).
- Single wire Half Duplex Communication

USART Features (3/3)

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- Multi-Processor communication
 - USART can enter Mute mode
 - Mute mode: disable receive interrupts
 - Wake up from mute mode (by idle line detection or address mark detection)
- Auto-baudrate detection using various character patterns.
- Driver enable (for RS485) signal sharing the same pin as nRTS.
- 14 interrupt sources

STM32F30x USART Implementation

USART features	USARTs1/2 /3	UART4	UART5
Hardware Flow Control	YES	NO	NO
Continuous communication using DMA	YES	YES	NO
Multiprocessor communication	YES	YES	YES
Synchronous mode	YES	NO	NO
Smartcard mode	YES	NO	NO
Single wire half duplex mode	YES	YES	YES
IrDA	YES	YES	YES
LIN	YES	YES	YES
Dual clock domain and wake up from STOP mode	YES	YES	YES
Receiver timeout	YES	YES	YES
Modbus Communication	YES	YES	YES
Autobaudrate detection	YES	YES	YES
Driver enable	YES	NO	NO

STM32F37x USART Implementation

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USART features	USART1	USART2	USART3
Hardware Flow Control	YES	YES	YES
Continuous communication using DMA	YES	YES	YES
Multiprocessor communication	YES	YES	YES
Synchronous mode	YES	YES	YES
Smartcard mode	YES	YES	YES
Single wire half duplex mode	YES	YES	YES
IrDA	YES	YES	YES
LIN	YES	YES	YES
Dual clock domain and wake up from STOP mode	YES	YES	YES
Receiver timeout	YES	YES	YES
Modbus Communication	YES	YES	YES
Autobaudrate detection	YES	YES	YES
Driver enable	YES	YES	YES

- When USART_CLK clock is HSI or LSE, the USART is able to wakeup the MCU from STOP.
- Wakeup from STOP is enabled by setting UESM bit in the USART_CR1.
- The sources of wake up from STOP mode can be the standard RXNE interrupt (the RXNEIE bit must be set before entering Stop mode). Or, a specific interrupt may be selected through the WUS bit fields in the USART_CR3:
 - Wake up on address match
 - Wake up on Start bit detection
 - Wake up on RXNE

How calculating the max baudrate allowing to wakeup properly from STOP

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- When $M = 0$, Max baudrate = $(DWU(max) * 10) / T(WU)$
- When $M = 1$, Max baudrate = $(DWU(max) * 11) / T(WU)$

With $T(WU)$ is the wakeup time from STOP mode

$DWU(max)$ is the receiver tolerance

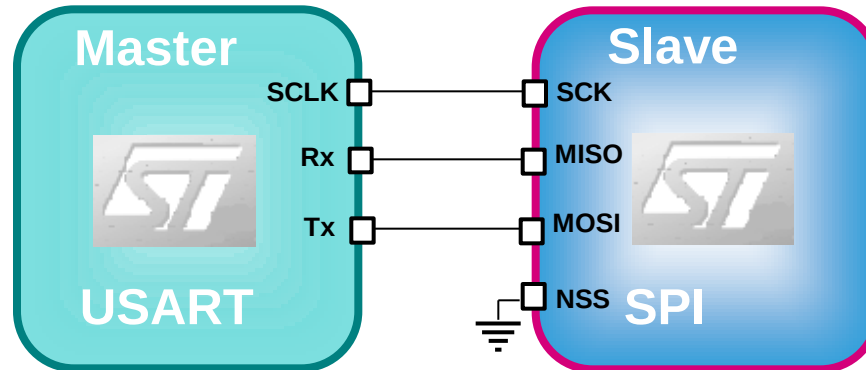
- Example: $M = 1$, $OVER8 = 0$, $ONEBIT = 0$,
→ Max baudrate = $(3.03\% * 11) / 4\mu s = 83,32 \text{ KBaud}$

- Each USART has DMA Tx and Rx requests
- Each of the USARTs (except UART5 in STM32F30x) requests is mapped on a different DMA channels: possibility to use DMA for all USARTs transfer direction at the same time.

Synchronous Mode

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- USART supports Full duplex synchronous communication mode
 - Full-duplex, three-wire synchronous transfer
 - USART Master mode only
 - Programmable clock polarity (CPOL) and phase (CPHA)
 - Programmable Last Bit Clock Pulse (LBCL) generation
 - Transmitter Clock output (SCLK)

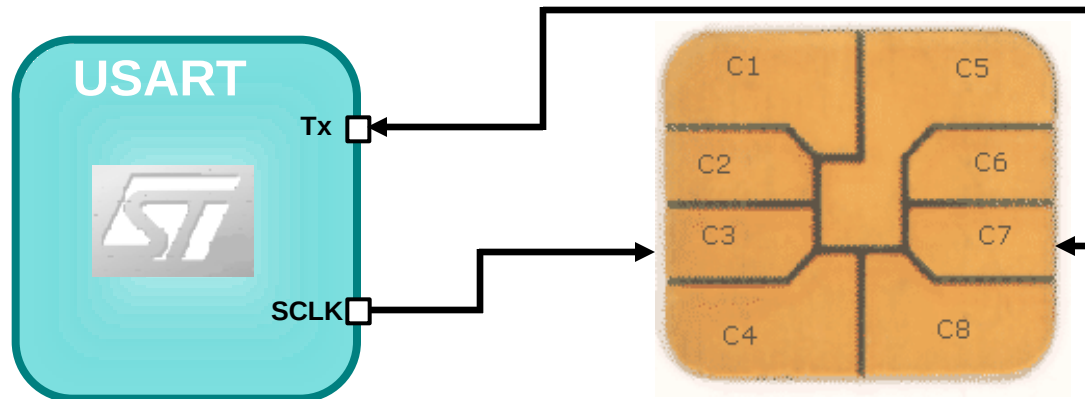


Full Duplex

Smart Card mode (1/2)

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- USART supports Smart Card Emulation ISO 7816-3
 - Half-Duplex, Clock Output (SCLK)
 - 9Bits data, 1.5 Stop Bits in transmit and receive.
 - T=0, T = 1 support
 - Programmable Clock Prescaler to guarantee a wide range clock input



Smart Card mode (2/2)

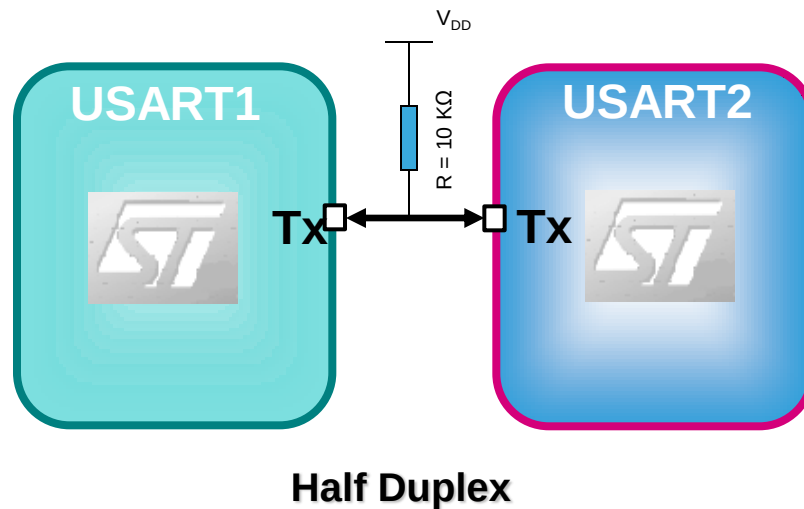
STM32F3 vs STM32F1xx

Features		STM32F3xx	STM32F1xx
Maximum USART baudrate in Smartcard mode		3Mbits/s	4.5Mbits/s
T = 0	In case of transmission error, according to the protocol specification, the USART should send the character.	The USART can handle automatic re-sending of data. The number of retries is programmable (8 max).	The data retry should be done by software.
	In case of reception error, according to the protocol specification, the smartcard must resend the same character.	The number of maximum retries is programmable (8 max). If the received character is still erroneous after the programmed number of retries, the USART will stop transmitting the NACK and will signal the error as a parity error.	
	A programmable guardtime is automatically inserted between two consecutive characters in transmission.	Yes	No
 T = 1	Character Wait Time (CWT)	Implemented using the new timeout feature.	All T = 1 feature should be implemented by software.
	Block Wait Time (BWT)		
	Block length and end of block detection		
	Direct/Inverse convention	Implemented using some new features: MSB/LSBFIRST, Binary data inversion etc...	

Single Wire Half Duplex mode

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- USART supports Half duplex synchronous communication mode
 - Only Tx pin is used (Rx is no longer used)
- Used to follow a single wire Half duplex protocol.



IrDA SIR Encoder Decoder

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- USART supports the IrDA Specifications
 - Half-duplex, NRZ modulation,
 - Max bit rate 115200 bps
 - The pulse width is 3/16 bit duration in normal mode

Modbus communication

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- Basic support for the implementation of Modbus/RTU and Modbus/ASCII protocols.
 - Modbus/RTU: In this mode, the end of one block is recognized by a “silence” (idle line) for more than 2 character times. This function is implemented through the programmable timeout function.
 - Modbus/ASCII: In this mode, the end of a block is recognized by a specific (CR/LF) character sequence. The USART manages this mechanism using the character match function.

Auto-baudrate detection

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- 4 patterns for auto-baudrate detection:
 - Any character starting with a bit at 1 → the USART will measure the duration of the START bit (Falling edge to rising edge).
 - Any character starting with a 10xx bit pattern → the USART will measure the duration of the START bit and of the first data bit (Falling edge to Falling edge).
 - A 0x7F character frame (it may be a 0x7F character in LSB first mode or a 0xFE in MSB first mode). In this case, the USART measures the duration of the start bit and the duration of bit 6.
 - A 0x55 character frame. In this case, the USART measures the duration of the start bit, the duration of bit 0 and the duration of bit 6. In parallel, another check is performed for each intermediate transition of RX line.
- Once the automatic baudrate detection is activated, the USART will wait for the first character on the RX line. The auto-baudrate completion is indicated by the setting of ABRF flag.
- The clock source frequency must be compatible with the expected communication speed : When oversampling by 16, the baud rate is between $f_{CK}/65535$ and $f_{CK}/16$. When oversampling by 8, the baudrate is between $f_{CK}/65535$ and $f_{CK}/8$.
- If the line is noisy, the correct baudrate detection is not guaranteed (BRR content may be corrupted))

USART Interrupts

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Interrupt event	Interrupt flag
Transmit Data Register Empty	TXE
Transmission Complete	TC
CTS	CTSIF
Receive Data Register Not Empty	RXNE
Overrun Error	ORE
Idle line detection	IDLE
Parity Error	PE
LIN break	LBDF
Noise Flag, Overrun error and Framing Error in multibuffer communication.	NE, ORE, FE
Character Match	CMF
Receiver timeout error	RTOF
End of Block	EOBF
Wakeup from STOP mode	WUF

STM32F1/F2/L USART vs STM32F3 USART

Main features (1/2)

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Feature	STM32F3	STM32F1/2/L
Programmable data length (8 or 9 bits)	Yes	Yes
Configurable stop bits	1, 1.5, 2	0.5, 1, 1.5, 2
Synchronous mode (Master only)	Yes	Yes
Single wire Half duplex	Yes	Yes
Programmable parity	Yes	Yes
Hardware flow control (nCTS/nRTS)	Yes	Yes
Driver Enable (for RS485)	Yes	No
Swappable Tx/Rx pin	Yes	No
IrDA	Yes	Yes
Basic support for Modbus	Yes	No
LIN	Yes	Yes
Smartcard	Yes (T = 0, T=1)	Yes (T = 0)
Dual Clock domain and wake up from STOP mode	Yes	No
Programmable data order with MSB first or LSB first	Yes	No

STM32F1/F2/L USART vs STM32F3 USART

Main features (2/2)

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Feature	STM32F3	STM32F1/2/L
Receiver timeout	Yes	No
Auto-baudrate detection	Yes	No
Continuous communication using DMA	Yes	Yes
Address/character match interrupt	Yes	No
End of Block interrupt	Yes	No
Multiprocessor communication	Yes	Yes

- How many USART interfaces are in the STM32F30x/F37x microcontrollers ?

- What is the maximum USART Baudrate?

- What are the features that are not supported by the UART4/5 in the STM32F30x?

- What are the different USART DMA requests ?



Thank you !

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